



SIES College of Arts, Science and Commerce (Autonomous) Sion (W),

Mumbai – 400 022.

CERTIFICATE

This is to certify that Mr. PRATHAM REDDY, Roll No. SMSC2526135 has successfully completed the necessary course of experiments in the subject of Next Generation Technologies during the academic year 2025-26, complying with the requirements for the course of M.Sc. Data Science [Semester-IV]

Prof. In-Charge

Head of the Department

Prof. Neha Ansari

Prof. Abuzar Ansari

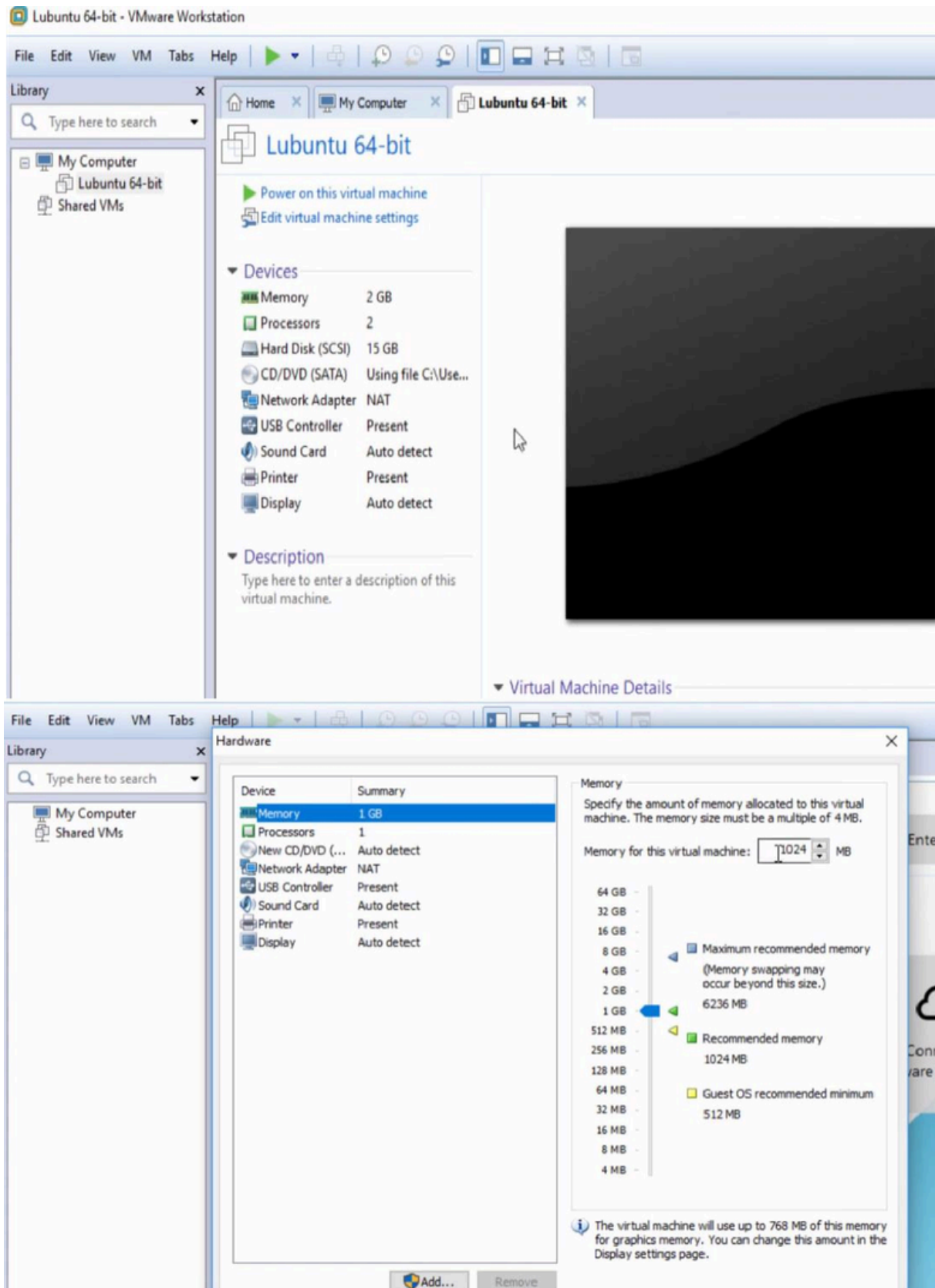
Examination Date:

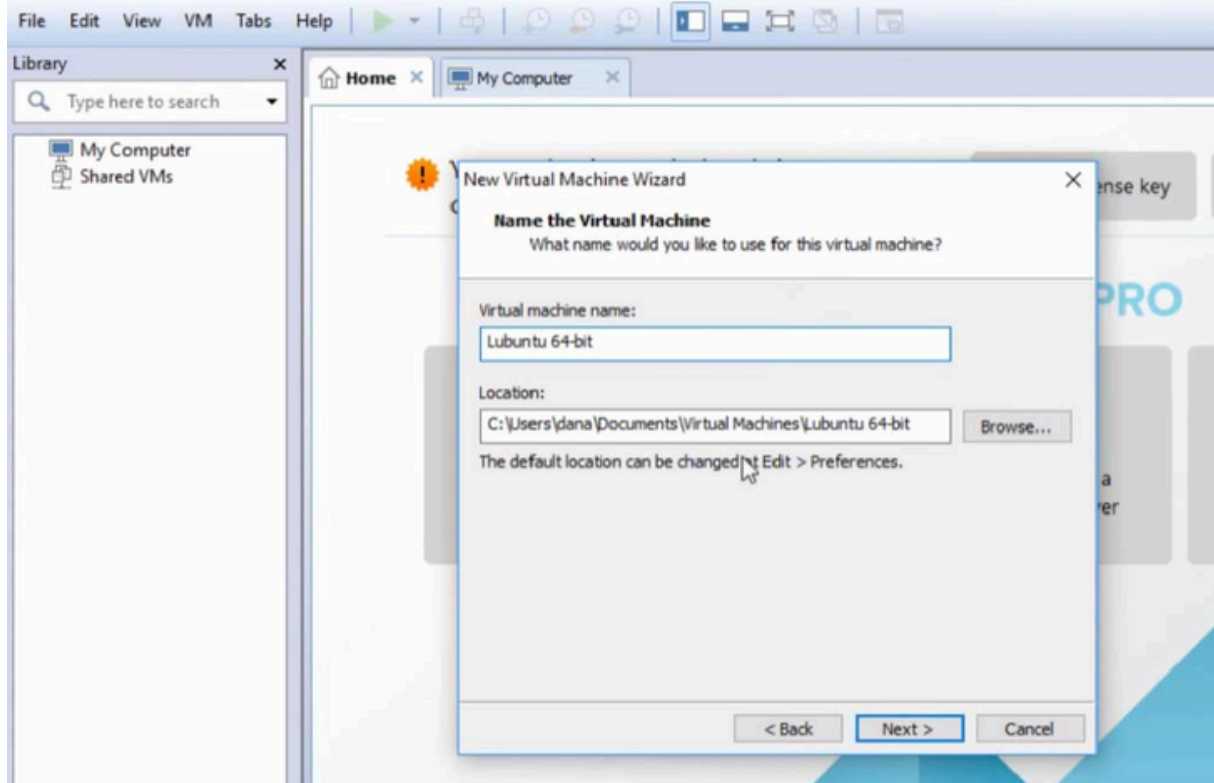
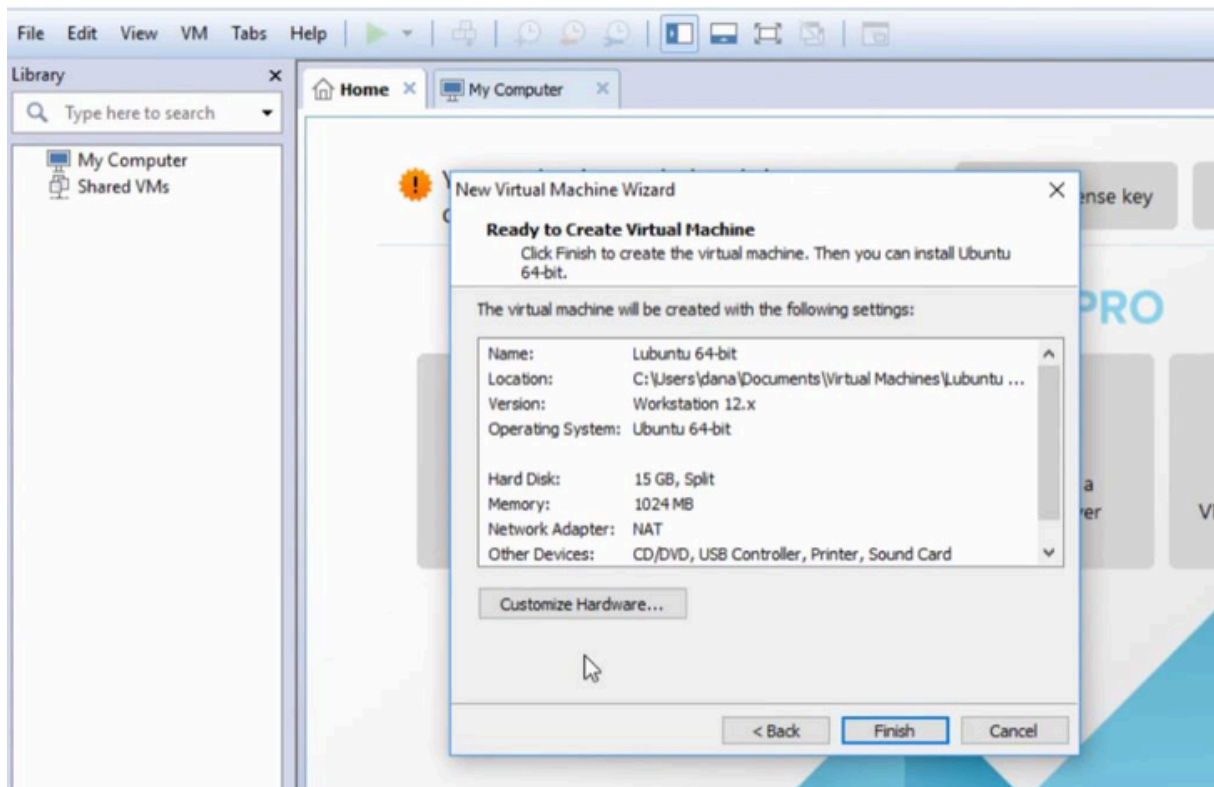
Examiner's Signature & Date: _____

College Seal & Date

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Practical 01: - Create Virtual Machine in Local Environment





Microsoft Azure

Upgrade

Search resources, services, and docs (G+J)

Copilot

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

(New) RG1

Create new

Instance details

Virtual machine name * ⓘ

linux-VM

Region * ⓘ

(Asia Pacific) Central India

Deploy to an Azure Extended Zone

Availability options ⓘ

Availability zone

Zone options ⓘ

☒ Self-selected zone

Choose up to 3 availability zones, one VM per zone

☐ Azure-selected zone (Preview)

Let Azure assign the best zone for your needs

Using an Azure-selected zone is not supported in region 'Central India'.

Availability zone * ⓘ

Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

< Previous

Next : Disks >

Review + create

File Edit View VM Tabs Help

Library

Type here to search

My Computer

Shared VMs

Home My Computer

New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

☐ Installer disc image file (iso):

C:\Users\dana\Desktop\New ISOs\linuxmint-18.2-ma

Browse...

☒ I will install the operating system later.

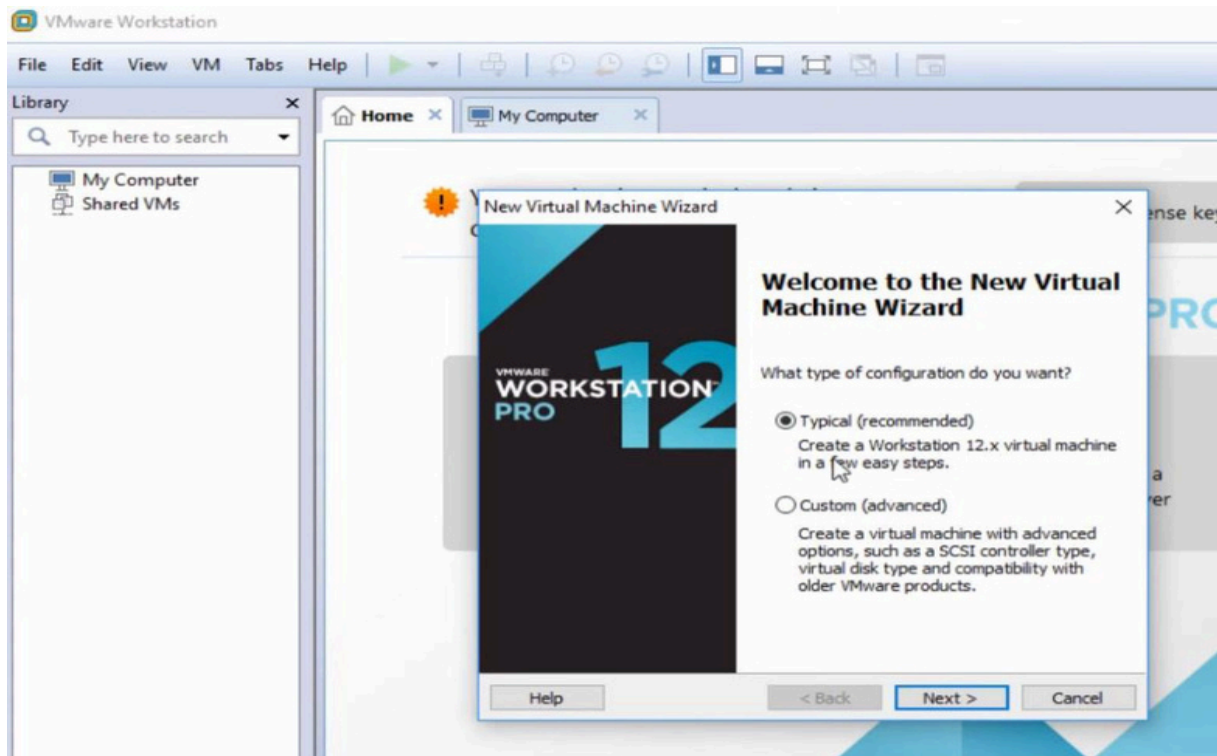
The virtual machine will be created with a blank hard disk.

Help

< Back

Next >

Cancel



Practical 02:- Introduction To Microsoft Azure

A. Creating a Virtual Machine with Azure Portal

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload

Help me create a low cost VM

Help me

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Standard HDD OS disks will be retired on September 8, 2028. [Click to learn more](#)

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption
Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

OS disk type *
The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Home > Compute infrastructure | Virtual machines

Create a virtual machine



Help me choose the right VM size for my workload

Help me create a low cost VM

OS disk

OS disk size ⓘ

Image default (30 GiB)

OS disk type * ⓘ

Standard HDD (locally-redundant storage)

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM ⓘ



Key management ⓘ

Platform-managed key

Enable Ultra Disk compatibility ⓘ



Data disks for linux

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM ⓘ
Create and attach a new disk Attach an existing disk					

Advanced

< Previous

Next : Networking >

Review + create

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network ⓘ	(New) vnet-centralindia (RG1)
	Edit virtual network
Subnet * ⓘ	(New) snet-centralindia-1
	Edit subnet 172.16.0.0 - 172.16.0.255 (256 addresses)
Public IP ⓘ	(new) linux-ip
	Create new
	Public IP addresses have a nominal charge. Estimate price
NIC network security group ⓘ	☐ None
	☒ Basic
	☐ Advanced
Public inbound ports * ⓘ	☐ None
	☒ Allow selected ports
Select inbound ports *	SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

☒ Allow selected ports

Select inbound ports *	SSH (22)
------------------------	----------

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Delete public IP and NIC when VM is deleted ⓘ

☒

Enable accelerated networking ⓘ

☐

The resource provider 'Microsoft.Network' should be registered in order to enable accelerated networking. [Learn more](#)

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options ⓘ	☒ None
	☐ Azure load balancer
	Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
	☐ Application gateway
	Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

Create a virtual machine

[Help me choose the right VM size for my workload](#)[Help me create a low cost VM](#)[Help me](#)Basics Disks Networking **Management** Monitoring Advanced Tags Review + create

Configure management options for your VM.

Microsoft Defender for Cloud

Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

Enable basic plan for free



This will apply to every VM in the selected subscription

Identity

Enable system assigned managed identity



Microsoft Entra ID

Login with Microsoft Entra ID



RBAC role assignment of Virtual Machine Administrator Login or Virtual Machine User Login is required when using Microsoft Entra ID login. [Learn more](#)

Microsoft Entra ID login now uses SSH certificate-based authentication. You will need to use an SSH client that supports OpenSSH certificates. You can use Azure CLI or Cloud Shell from the Azure Portal. [Learn more](#)

Auto-shutdown

Enable auto-shutdown



< Previous

Next : Monitoring >

Review + create

Create a virtual machine

[Help me choose the right VM size for my workload](#)[Help me create a low cost VM](#)[Help me](#)

User Login is required when using Microsoft Entra ID login. [Learn more](#)

Microsoft Entra ID login now uses SSH certificate-based authentication. You will need to use an SSH client that supports OpenSSH certificates. You can use Azure CLI or Cloud Shell from the Azure Portal. [Learn more](#)

Auto-shutdown

Enable auto-shutdown



Shutdown time

19:00:00

Time zone

(UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi

Notification before shutdown



Email *

kamalkishorfmsc2425139@ascs.sies.edu.in

Backup

Enable backup



Guest OS updates

Enable periodic assessment



Patch orchestration options

Image default

Some patch orchestration options are not available for this image. [Learn more](#)

< Previous

Next : Monitoring >

Review + create

Create a virtual machine

...



Help me choose the right VM size for my workload

Help me create

Basics

Disks

Networking

Management

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Tags

Review + create

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ⓘ

☐

Diagnostics

Boot diagnostics ⓘ

☒

Enable with managed storage account (recommended)

☐

Enable with custom storage account

☐

Disable



There is a nominal charge for the managed storage account.

Enable OS guest diagnostics ⓘ

☐

Health

Enable application health monitoring ⓘ

☐

< Previous

Next : Advanced >

Review + create

Create a virtual machine

[Help me choose the right VM size for my workload](#)[Help me create a low cost VM](#)[Help me](#)

Extensions ⓘ

[Select an extension to install](#)

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#)

[Select a VM application to install](#)

Custom data and cloud init

Pass a cloud-init script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

Custom data

Custom data on the selected image will be processed by cloud-init.
[Learn more about custom data for VMs](#)

User data

Pass a script, configuration file, or other data that will be accessible to your applications throughout the lifetime of the virtual machine. Don't use user data for storing your secrets or passwords. [Learn more about user data for VMs](#)

Create a virtual machine

[Help me choose the right VM size for my workload](#)[Help me create a low cost VM](#)

Performance (NVMe)

Enable capabilities to enhance the performance of your resources.

Higher remote disk storage performance with NVMe ⓘ

☐

The selected size is not supported for NVMe. [See supported size families](#)

Host

Azure Dedicated Hosts allow you to provision and manage a physical server within our data centers that are dedicated to your Azure subscription. A dedicated host gives you assurance that only VMs from your subscription are on the host, flexibility to choose VMs from your subscription that will be provisioned on the host, and the control of platform maintenance at the level of the host. [Learn more](#)

Host group ⓘ

No host groups found



Capacity reservations

Capacity reservations allow you to reserve capacity for your virtual machine needs. You get the same SLA as normal virtual machines with the security of reserving the capacity ahead of time. [Learn more](#)

Capacity reservations ⓘ

☒

None
Do not use reserved capacity

☐

Capacity reservation group
Use reserved capacity from this subscription

☐

(Preview) Shared capacity reservation group
Use reserved capacity from another subscription

Create a virtual machine

...

Help me choose the right VM size for my workload

Help me create a low cost VM

Help me create

✓ Validation passed

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the [Azure Marketplace Terms](#) for additional details.

Name

KAMALKISHOR.SUNWASIYA

Preferred e-mail address

kamalkishorfmsc2425139@asc.sies.edu.in

Preferred phone number

022-27708376

⚠

You have set SSH port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.

Basics

Subscription	Azure subscription 1
Resource group	RG1
Virtual machine name	linux
Region	Central India
Availability options	Availability zone

< Previous

Next >

Create

Home > Compute infrastructure | Virtual machines

Create a virtual machine

...

Help me choose the right VM size for my workload

Help me create a low cost VM

H

✓ Validation passed

Basics

Subscription	Azure subscription 1
Resource group	RG1
Virtual machine name	linux
Region	Central India
Availability options	Availability zone
Zone options	Self-selected zone
Availability zone	1
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No
Image	Ubuntu Server 24.04 LTS - Gen2
VM architecture	x64
Size	Standard B2ats v2 (2 vcpus, 1 GiB memory)
Enable Hibernation	No
Authentication type	Password
Username	azureuser

Microsoft Azure

Upgrade

Search resources, services, and docs (G+/I)

Copilot

kamalkishorfmisc242513...
SOUTH INDIAN EDUCATION SOC...

Home > Compute infrastructure | Virtual machines

Create a virtual machine

Help me choose the right VM size for my workload Help me create a low cost VM Help me create a VM optimized for high availability

Validation passed

Backup

Disabled

Enable periodic assessment

OFF

Enable hotpatch

OFF

Patch orchestration options

Image Default

Monitoring

Alerts

OFF

Boot diagnostics

On

Enable OS guest diagnostics

OFF

Enable application health monitoring

OFF

Advanced

Extensions

None

VM applications

None

Cloud init

No

User data

No

Disk controller type

SCSI

Proximity placement group

None

Capacity reservation group

None

Previous

Next

Create

Activate Windows
Go to Settings to activate Windows.
Download a template for automation Give feedback

Home

CreateVm-canonical.ubuntu-24_04-lts-server-20260205112453 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2...

Start time: 05/02/2026, 11:33:40

Subscription: Azure subscription 1

Correlation ID: a6c1cb5e-2d86-4f30-9f43-6453f5b69d136

Resource group: RG1

Deployment details

Resource	Type	Status	Operation details
linux113_21	Microsoft.Network/networkInterfaces	Created	Operation details
linux-nsg	Microsoft.Network/networkSecurityG...	OK	Operation details
linux-ip	Microsoft.Network/publicIPAddresses	OK	Operation details
network-interface-associated-virtual-m...	Microsoft.Resources/deployments	OK	Operation details

Give feedback

Tell us about your experience with deployment

Microsoft Defender for Cloud

Secure your apps and infrastructure

Go to Microsoft Defender for Cloud >

Free Microsoft tutorials

Start learning today >

Work with an expert

Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.

Find an Azure expert >

Activate Windows
Go to Settings to activate Windows.

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > CreateVm-canonical.ubuntu-24_04-its-server-20260205112453 | Overview

linux Virtual machine

Help me copy this VM in any region Manage this VM with Azure CLI

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Automation

Help

Standard HDD OS disks will be retired on September 8, 2026. →

Help me copy this VM in any region

Connect Start Restart Stop Hibernate Capture Delete Refresh Scale Open in mobile Feedback CU / PS

Essentials

Resource group (move) : RG1

Status : Running

Location : Central India (Zone 1)

Subscription (move) : Azure subscription 1

Subscription ID : 313cc70f-9b06-4dfc-9233-22e750991e6

Availability zone : 1

Operating system : Linux (ubuntu 24.04)

Size : Standard B2ats v2 (2 vcpus, 1 GiB memory)

Primary NIC public IP : 20.244.14.32

1 associated public IPs

Virtual network/subnet : vnet-centralindia/vnet-centralindia-1

DNS name : Not configured

Health state : *

Time created : 05/02/2026, 06:04 UTC

Tags (edit) : Add tags

Properties Monitoring Capabilities (7) Recommendations Tutorials

Virtual machine

Computer name : linux

Operating system : Linux (ubuntu 24.04)

VM generation : V2

Networking

Public IP address : 20.244.14.32 (Network interface linux13321)

1 associated public IPs

Public IP address (IPv6) : +

JSON View

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > CreateVm-canonical.ubuntu-24_04-its-server-20260205112453 | Overview > linux

linux | Connect Virtual machine

Search

Overview

Activity log

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Connect

Bastion

Networking

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Now view, configure, and even save your connection settings — all in one place. Have comments or suggestions for our new Connect experience? Provide feedback

Refresh Reset password or keys Manage JIT Troubleshoot Feedback

Native SSH

MOST POPULAR LOCAL MACHINE

Source machine

Source machine OS : Windows

Source IP address : Local IP | 183.87.170.248

Connecting over a VPN?

Destination VM

VM IP address : Public IP | 20.244.14.32

VM port : 22

Connection prerequisites

VM access : Check inbound NSG rules

Check access

SSH command

Execute in your choice of local shell

ssh azureuser@20.244.14.32

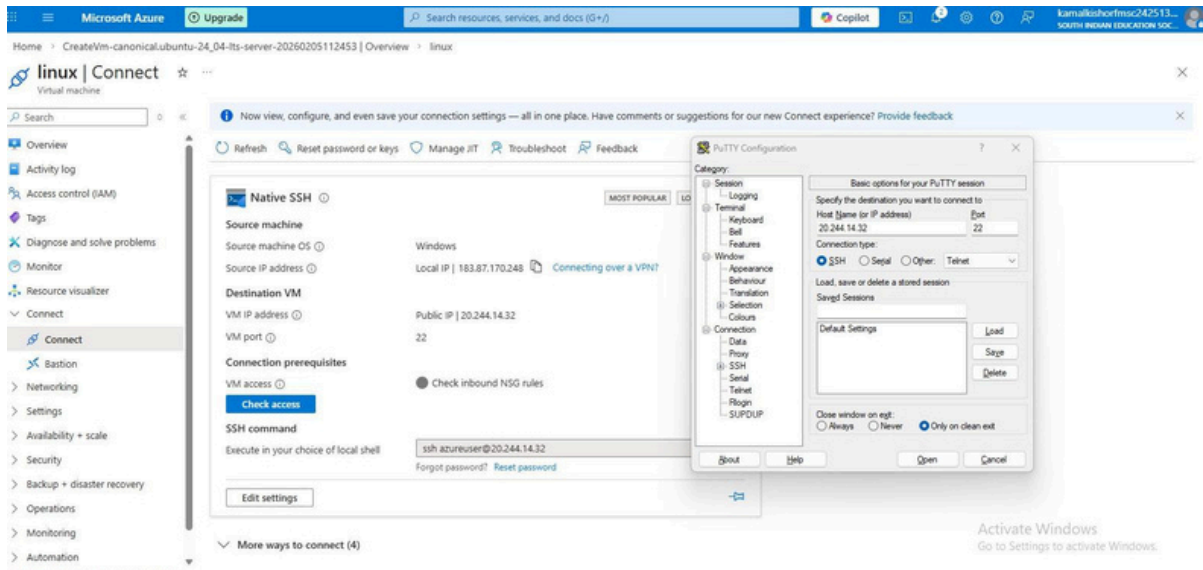
Forgot password? Reset password

Edit settings

More ways to connect (4)

Activate V Go to Setting

Add or remove favorites by pressing Ctrl+Shift+F



```
root@linux: /home/azureuser/koteshh
login as: azureuser
azureuser@20.244.14.32's password:
welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1017-azure x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Thu Feb  5 06:15:49 UTC 2026

System load:  0.08          Processes:            116
Usage of /:   5.6% of 28.02GB Users logged in:             0
Memory usage: 30%          IPv4 address for eth0: 172.16.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

updates can be applied immediately.

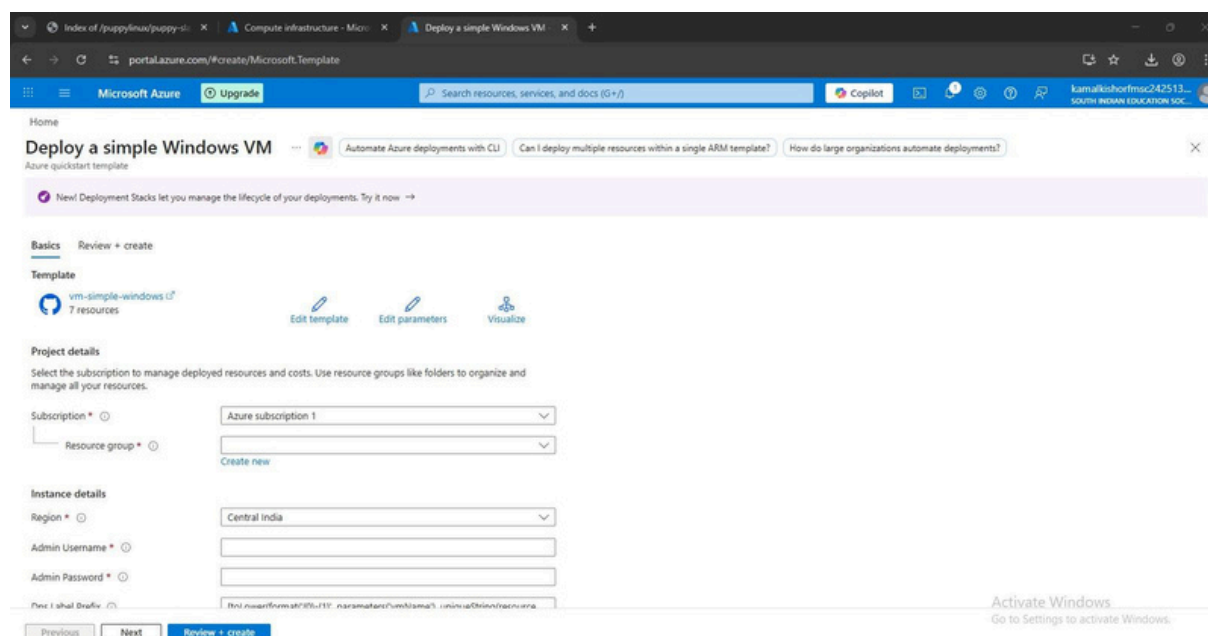
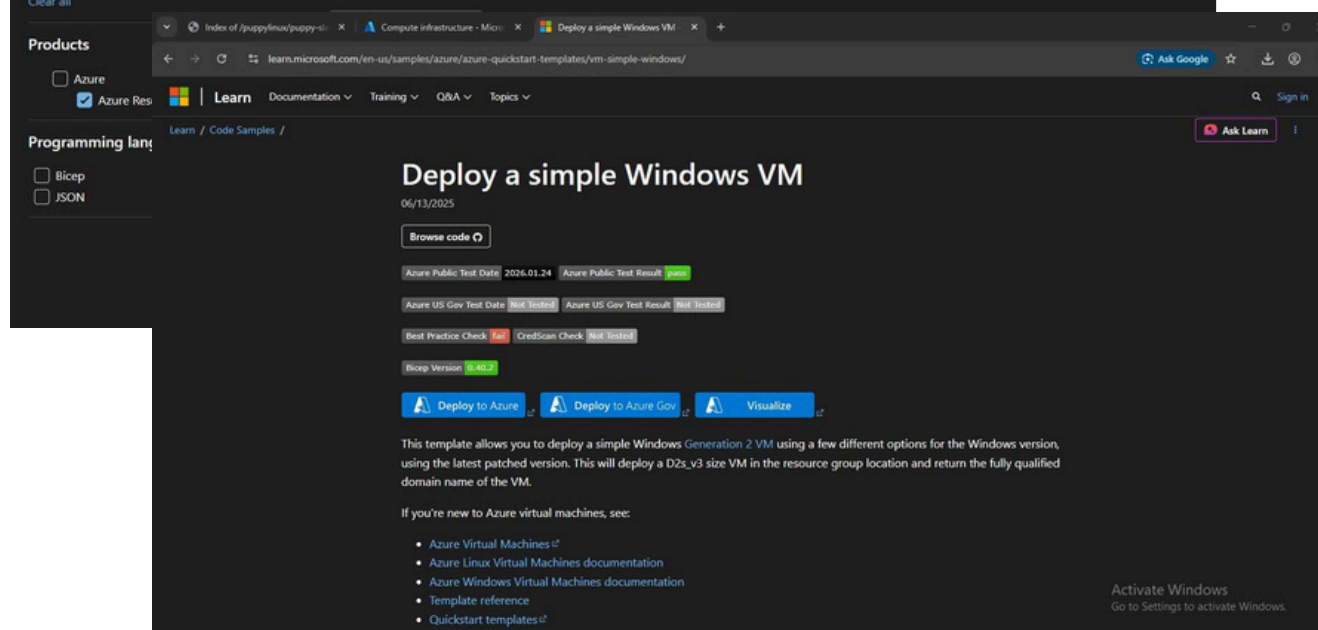
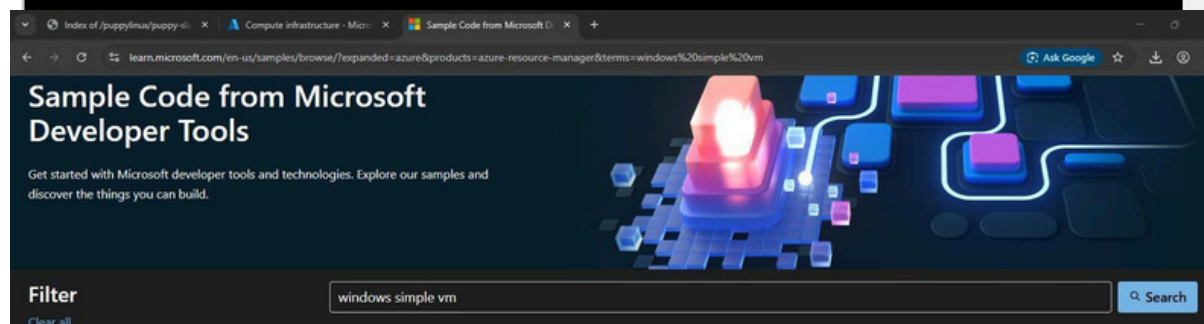
enable ESM Apps to receive additional future security updates.
see https://ubuntu.com/esm or run: sudo pro status
```



```
root@linux: /home/azureuser/kotesh
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

B. Creating a Virtual Machine using Template



Home > Deploy a simple Windows VM

Edit template

Edit your Azure Resource Manager template

+ Add resource ↑ Quickstart template ↑ Load file ↓ Download

adminUsername (securestring)	92	"location": {
	93	"type": "string",
dnsLabelPrefix (string)	94	"defaultValue": "[resourceGroup().location]",
	95	"metadata": {
publicIpName (string)	96	"description": "Location for all resources."
	97	}
publicIpAllocationMethod (string)	98	},
	99	"vmName": {
publicIpSku (string)	100	"type": "string",
	101	"defaultValue": "my-vm01",
OSVersion (string)	102	"metadata": {
	103	"description": "Name of the virtual machine."
vmSize (string)	104	}
	105	},
location (string)	106	"securityType": {
	107	"type": "string",
	108	"defaultValue": "TrustedLaunch",
vmName (string)	109	"allowedValues": [
	110	"Standard",
	111	"TrustedLaunch"
securityType (string)	112	],

Variables (13)

Resources (7)

Save

Discard

Microsoft Azure Upgrade Search resources, services, and docs (G+/I) Copilot kamalishorfmisc242513... SOUTH INDIAN EDUCATION SOC.

Home

Custom deployment

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1
Resource group * RG1
Create new

Instance details

Region * (Asia Pacific) Central India ✓
Admin Username * azureuser ✓
Admin Password * ***** ✓
Dns Label Prefix [toLower(format('{0}-{1}', parameters('vmName'), uniqueString(resource...)) ✓
Public Ip Name myPublicIP ✓
Public IP Allocation Method Static ✓
Public Ip Sku Standard ✓
OS Version 2022-datacenter-azure-edition ✓
Vm Size Standard D2s v5 ✓

Previous Next Review + create

Activate Windows
Go to Settings to activate Windows.

Microsoft Azure

Upgrade

Search resources, services, and docs (G+J)

Copilot

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Home

Custom deployment

Deploy from a custom template

as part of this template. Prices and associated legal terms for any Marketplace offerings can be found in the Azure Marketplace; both are subject to change at any time prior to deployment.

Neither subscription credits nor monetary commitment funds may be used to purchase non-Microsoft offerings. These purchases are billed separately.

If any Microsoft products are included in a Marketplace offering (e.g. Windows Server or SQL Server), such products are licensed by Microsoft and not by any third party.

Basics

Subscription

Resource group

Region

Admin Username

Admin Password

Dns Label Prefix

Public Ip Name

Public Ip Allocation Method

Public Ip Sku

OS Version

Vm Size

Location

Vm Name

Security Type

Azure subscription 1

RG1

Central India

azureuser

[toLower(format(0-1), parameters('vmName'), uniqueString(resourceGrou...

myPublicIP

Static

Standard

2022-datacenter-azure-edition

Standard_D2s_v5

[resourceGroup.location]

my-vm01

TrustedLaunch

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Activate Windows

Go to Settings to activate Windows.

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Search resources, services, and docs (G+J)

Copilot

kamalkishorfmisc242513... SOUTH INDIAN EDUCATION SOC...

Home > Microsoft.Template-20260205121206 | Overview > RG1

my-vm01

Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

Search

Help me copy this VM in any region

Connect

Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Open in mobile

Feedback

CU / PS

Overview

Activity log

Access control (IAM)

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Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Windows Admin Center

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Monitoring

Add or remove favorites by pressing Ctrl + Shift + F

Essentials

Resource group (move)

Status

Location

Subscription (move)

Subscription ID

Operating system

Size

Primary NIC public IP

Virtual network/subnet

DNS name

Health state

Time created

Tags (edit)

Add tags

Properties

Monitoring

Capabilities (8)

Recommendations

Tutorials

Virtual machine

Computer name

Operating system

VM generation

Agent status

Agent version

my-vm01

Windows (Windows Server 2022 Datacenter Azure Edition)

V2

Ready

2.7.41491.1183

Networking

Public IP address

Public IP address (IPv6)

Private IP address

Private IP address (IPv6)

4.224.123.119 (Network interface myVMNIC)

1 associated public IPs

0.0.0.4

0.0.0.0

Activate Windows

Go to Settings to activate Windows.

JSON View

Microsoft Azure

Upgrade

Search resources, services, and docs (G+J)

Copilot

kamalkishorfmisc242513... SOUTH INDIAN EDUCATION SOC...

Home > Microsoft.Template-20260205121206 | Overview > RG1 > my-vm01

RG1

Resource group

How do I monitor this resource group?

Are there any alerts fired for this resource group?

What are the best practices for managing this resource group?

Search

Create

Manage view

Delete resource group

Refresh

Export to CSV

Open query

Assign tags

Move

Delete

Group by none

Overview

Activity log

Access control (IAM)

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Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Resources

Recommendations

Filter for any field...

Type equals all

Location equals all

Add filter

Name	Type	Location
bootdiagdpvprncpsla	Storage account	Central India
default-NSG	Network security group	Central India
linux-nsg	Network security group	Central India
my-vm01	Virtual machine	Central India
my-vm01_disk2_e6b65ec3bfa4cfa0399b0ad2d8bb53	Disk	Central India
my-vm01_OsDisk_1_7eec1532717848338dabf966442ba38	Disk	Central India
myPublicIP	Public IP address	Central India
myVMNIC	Network interface	Central India
MyVNET	Virtual network	Central India
vnet-centralindia	Virtual network	Central India

Showing 1 - 10 of 10. Display count: auto

Activate Windows

Go to Settings to activate Windows.

Give feedback

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Microsoft.Template-20260205121206 | Overview > RG1 > my-vm01

RG1 Resource group

How do I monitor this resource group? Are there any alerts fired for this resource group?

Search

Create Manage view Delete resource group Refresh

Overview

Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Monitoring Automation Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all

Name ?

- bootdiagqdpvrncspla
- default-NSG
- GuestAttestation [my-vm01/GuestAttestation]
- linux-nsG
- my-vm01
- my-vm01_disk2_e8b65ec63bfa4cfa0399b0ad2d8bb53
- my-vm01_OsDisk_1_7eec1532717848338dabff966442ba38
- myPublicIP
- myVMNIC
- MyVNET
- vnet-centralindia

Showing 1 - 10 of 10. Display count: auto

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

RG1

11 resources being deleted across 1 subscriptions, 1 resource groups

All dependent resources, including hidden types, are shown

Name	Resource type	Subscription	Resource Group
bootdiagqdpvrncspla	Storage account	Azure subscription 1	RG1
default-NSG	Network security group	Azure subscription 1	RG1
GuestAttestation [my-vm01/GuestAttestation]	Microsoft.Compute/virtualMachines/...	Azure subscription 1	RG1
linux-nsG	Network security group	Azure subscription 1	RG1
my-vm01	Virtual machine	Azure subscription 1	RG1
my-vm01_disk2_e8b65ec63bfa4cfa0399b0ad2d8	Disk	Azure subscription 1	RG1
my-vm01_OsDisk_1_7eec1532717848338dabff966	Disk	Azure subscription 1	RG1
myPublicIP	Public IP address	Azure subscription 1	RG1

Apply force delete for selected Virtual machines and virtual machine scale sets

Enter resource group name to confirm deletion *

RG1

Delete Cancel

Activate Windows
Go to Settings to activate Windows.

Microsoft Azure | Upgrade | Search resources, services, and docs (G+)

Home > Microsoft.Template-20260205121206 | Overview > RG1 > my-vm01

RG1 Resource group

How do I troubleshoot issues with this resource group? How do I monitor this resource group? Are there any alerts fired for this resource group?

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete

Group by none

Overview

Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Monitoring Automation Help

Essentials

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name ?	Type	Location
default-NSG	Network security group	Central India
my-vm01_disk2_e8b65ec63bfa4cfa0399b0ad2d8bb53	Disk	Central India
my-vm01_OsDisk_1_7eec1532717848338dabff966442ba38	Disk	Central India
myPublicIP	Public IP address	Central India
myVMNIC	Network interface	Central India
MyVNET	Virtual network	Central India

Showing 1 - 6 of 6. Display count: auto

Activate Windows
Go to Settings to activate Windows.
Give feedback

Practical 03:- Configuring and Managing Virtual Machines in Azure

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

i This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

RG1

[Create new](#)

Instance details

Virtual machine name * ⓘ

Region * ⓘ

(Asia Pacific) Central India

[Deploy to an Azure Extended Zone](#)

Availability options ⓘ

No infrastructure redundancy required

Security type ⓘ

Trusted launch virtual machines

[Configure security features](#)

Image * ⓘ

 Ubuntu Server 24.04 LTS - x64 Gen2 (free services eligible)

[See all images](#) | [Configure VM generation](#)

Size * ⓘ

Standard_DC1ds_v3 - 1 vcpu, 8 GiB memory (\$87.60)

[See all sizes](#)

Enable Hibernation ⓘ

☐

i Hibernate does not currently support Trusted launch and Confidential virtual machines for Linux images. [Learn more](#)

Administrator account

Authentication type ⓘ

☐ SSH public key
 ☒ Password

Username * ⓘ

Sakshi

Password *

.....

Confirm password *

.....

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☐ None
 ☒ Allow selected ports

Select inbound ports *

SSH (22)

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

```

root@VM1: /home/Hussain

Memory usage: 5%      IPv4 address for eth0: 10.10.10.2
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

Hussain@VM1:~$ sudo su
root@VM1:/home/Hussain#
    
```

Practical 04:- Scaling Virtual Machine

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 ☐ JSON

Azure Resource Manager

14 results for "deploy simple window"

Deploy a simple Windows VM

06/11/2023

This template allows you to deploy a simple Windows VM using a few different options for the Windows version, using the latest patched version. This will deploy an A2 size...

Azure Bicep

Add

Deploy a simple Windows VM with monitoring and diagnostics

07/06/2022

This template allows you to deploy a simple Windows VM along with the diagnostics extension which enables monitoring and diagnostics for the VM

Azure JSON

Add

Deploy a simple VM Scale Set with Windows VMs

07/06/2022

This template allows you to deploy a simple VM Scale Set of Windows VMs using the latest patched version of various Windows Versions. These VMs are behind a load...

Azure JSON

Add

Deploy a simple Windows VM with tags

06/11/2023

This template will deploy a D2_v3 Windows VM, NIC, Storage Account, Virtual Network, Public IP Address, and Network Security Group. The tag object is created in the...

Azure Bicep

Add

Deploy a simple VM Scale Set with Windows VMs and a Jumpbox

07/06/2022

This template allows you to deploy a simple

Deploy a Windows VM and configures WinRM https listener

07/06/2022

This template allows you to deploy a simple

Deploy a VM Scale Set with a Windows custom image

07/06/2022

This template allows you to deploy a simple

Deploy a Windows VM scale set with Azure Application Gateway

07/16/2023

This template allows you to deploy a simple

Deploy a simple Windows VM

06/13/2025

[Browse code](#)

Azure Public Test Date 2026.01.24 Azure Public Test Result pass

Azure US Gov Test Date Not Tested Azure US Gov Test Result Not Tested

Best Practice Check fail CredScan Check Not Tested

Bicep Version 0.40.2



Deploy to Azure



Deploy to Azure Gov



Visualize



This template allows you to deploy a simple Windows [Generation 2 VM](#) using a few different options for the Windows version, using the latest patched version. This will deploy a D2s_v3 size VM in the resource group location and return the fully qualified domain name of the VM.

If you're new to Azure virtual machines, see:

[+ Add resource](#) [↑ Quickstart template](#) [↑ Load file](#) [↓ Download](#)

Parameters (11)

Variables (13)

Resources (7)

```
11  "parameters": {
53    "OSVersion": {
81      "metadata": {
83      }
84    },
85    "vmSize": {
86      "type": "string",
87      "defaultValue": "Standard_D2s_v5",
88      "metadata": {
89        "description": "Size of the virtual machine."
90      }
91    },
92    "location": {
93      "type": "string",
94      "defaultValue": "[resourceGroup().location]",
95      "metadata": {
96        "description": "Location for all resources."
97      }
98    },
99    "vmName": {
100      "type": "string",
101      "defaultValue": "Sakshi-vm",
102      "metadata": {
103        "description": "Name of the virtual machine."
104      }
105    },
106    "securityType": {
107      "type": "string",
108      "defaultValue": "TrustedLaunch",
109      "allowedValues": [
110        "Standard",
111        "TrustedLaunch"
112      ],
```

Save

Discard

Template

 Customized template 
7 resources

 Edit template

 Edit parameters

 Visualize

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Azure subscription 1

Resource group * ⓘ

[Create new](#)

Instance details

Region * ⓘ

Admin Username * ⓘ

Admin Password * ⓘ

Dns Label Prefix ⓘ

Public Ip Name ⓘ

Public IP Allocation Method ⓘ

Public Ip Sku ⓘ

OS Version ⓘ

A resource group is a container that holds related resources for an Azure solution.

Name *

RG5

OK

Cancel

myPublicIP

Static

Standard

2022-datacenter-azure-edition

Microsoft.Template-20260205130101 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

*** Deployment is in progress

Deployment name : Microsoft.Template-20260205130101

Subscription : Azure subscription 1

Resource group : RG5

Start time : 2/5/2026, 12:59:48 PM

Correlation ID : 70bd9f3c-87d8-4e87-aeb3-43c345cdf0f0

Deployment details

Resource	Type	Status	Operation details
Sakshi-vm/GuestAttestation	Microsoft.Compute/virtualMach	Created	Operation details
Sakshi-vm	Virtual machine	OK	Operation details
bootdiags7emmbcqng6mk	Storage account	OK	Operation details
myVMNIC	Microsoft.Network/networkinter	OK	Operation details
myPublicIP	Public IP address	OK	Operation details
MyVNET	Virtual network	OK	Operation details
bootdiags7emmbcqng6mk	Storage account	OK	Operation details
myPublicIP	Public IP address	OK	Operation details
default-NSG	Network security group	OK	Operation details

Give feedback

Tell us about your experience with deployment

RG5

Are there any alerts fired for this resource group? What are the best practices for managing this resource group? How do I monitor this resource group?

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Export template Open in mobile

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Subscription (move)

Azure subscription 1

Deployments

1 Succeeded

Subscription ID

dcdb0dbd-4d08-4169-971a-c6e233d3235b

Location

Central India

Tags (edit)

Add tags

Resources

Recommendations

Filter for any field... Type equals all Location equals all Add filter

Name 1	Type	Location
bootdiags7emmbcqng6mk	Storage account	Central India
default-NSG	Network security group	Central India
myPublicIP	Public IP address	Central India
myVMNIC	Network interface	Central India
MyVNET	Virtual network	Central India
SAKSHI-VM	Virtual machine	Central India
Sakshi-vm_disk2_8fda3c50a0434b57bd99d80b083340f6	Disk	Central India
Sakshi-vm_OsDisk_1_c85ad3b69b724477b5ed976a48cd5c06	Disk	Central India

RG5

Are there any alerts fired for this resource group? What are the best practices for managing this resource group? How do I monitor this resource group?

Search

Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags Move Delete Export template Open in mobile

Overview

Activity log

Access control (IAM)

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Essentials

Subscription (move)

Azure subscription 1

Deployments

1 Succeeded

Subscription ID

dcdb0dbd-4d08-4169-971a-c6e233d3235b

Location

Central India

Tags (edit)

Add tags

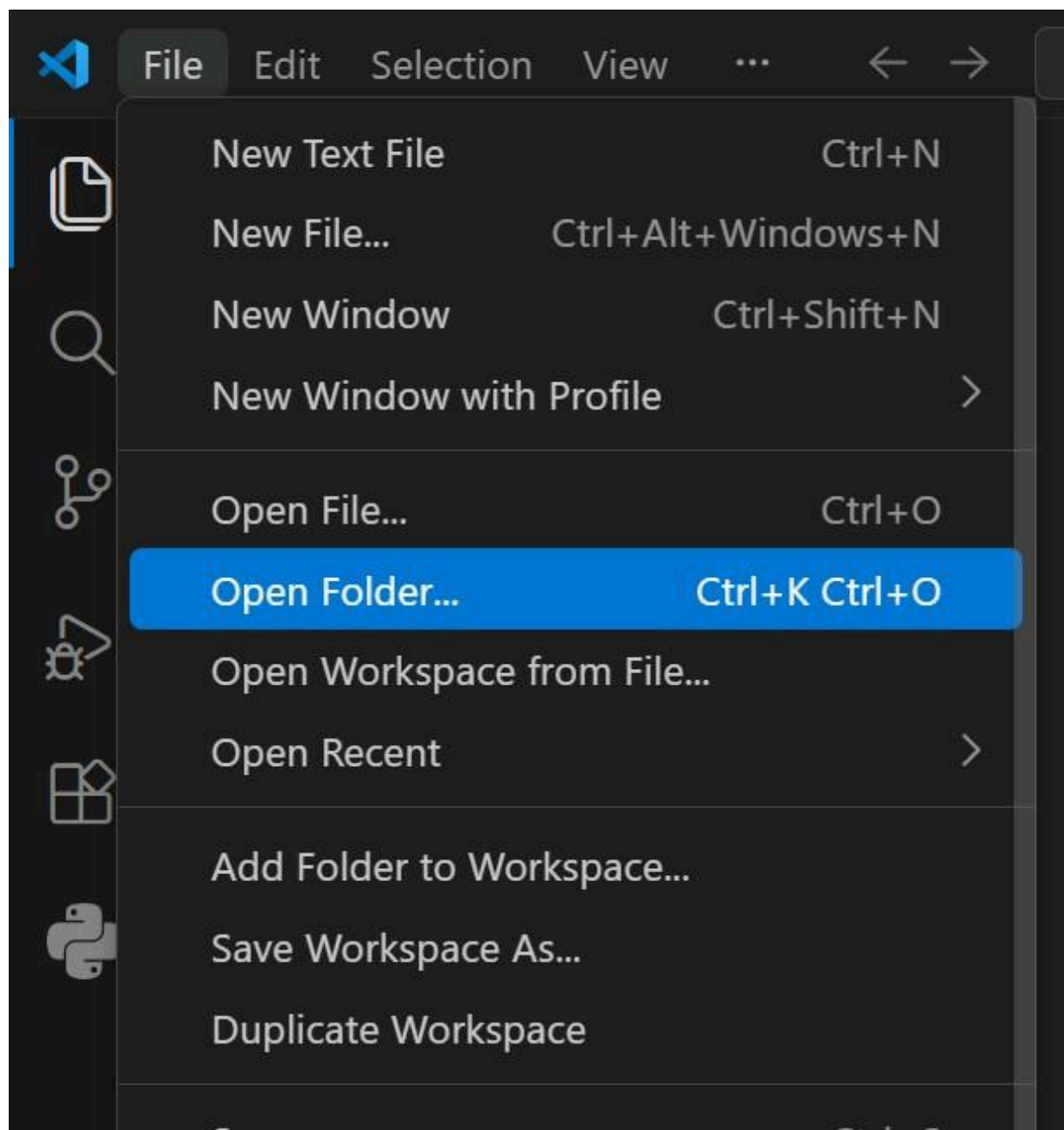
Resources

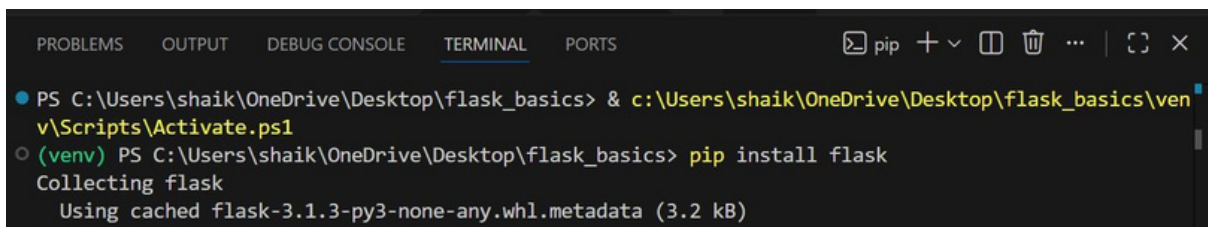
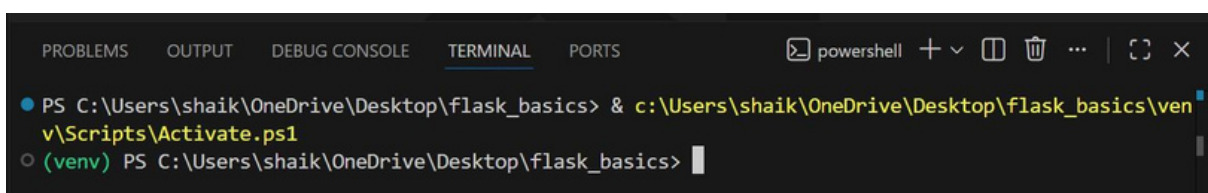
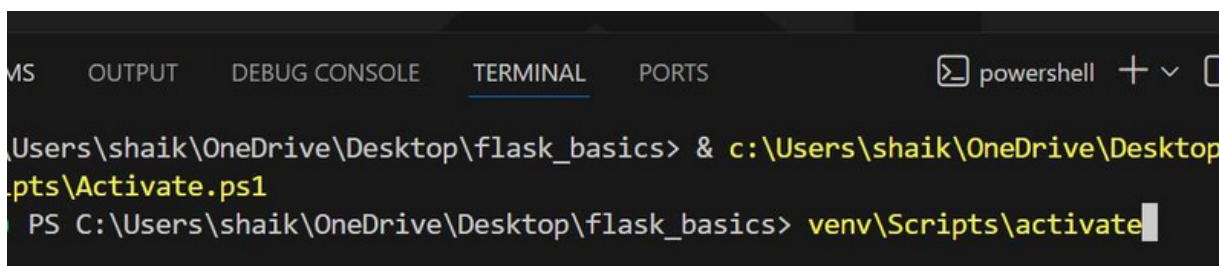
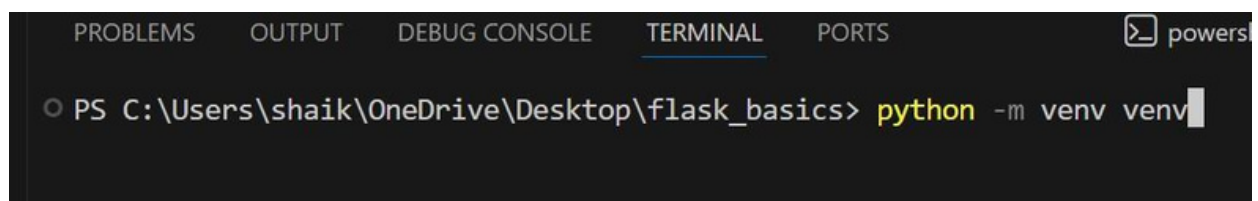
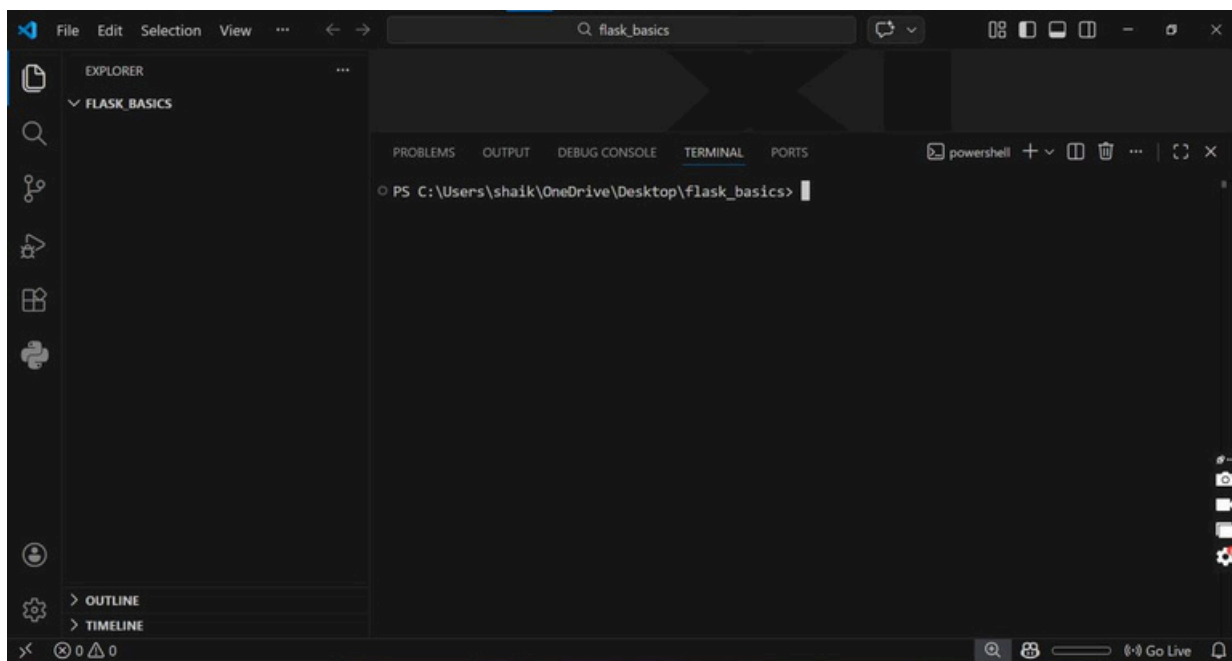
Recommendations

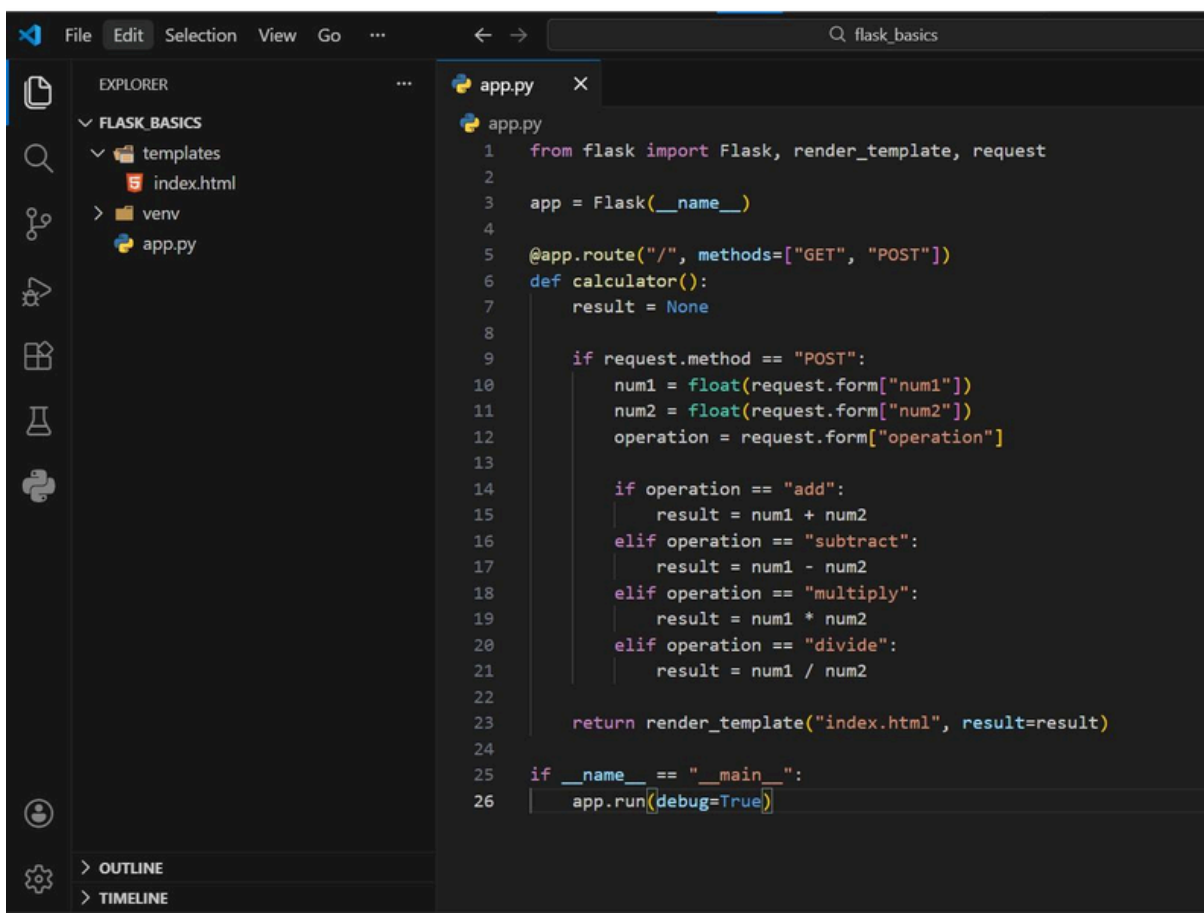
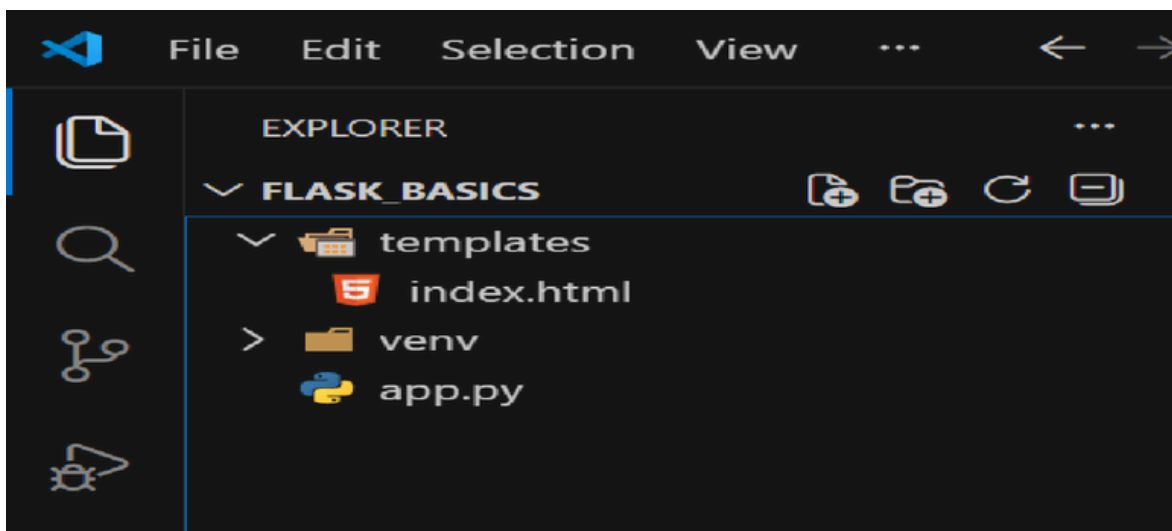
Filter for any field... Type equals all Location equals all Add filter

Name 1	Type	Location
bootdiags7emmbcqng6mk	Storage account	Central India
default-NSG	Network security group	Central India
myPublicIP	Public IP address	Central India
myVMNIC	Network interface	Central India
MyVNET	Virtual network	Central India
SAKSHI-VM	Virtual machine	Central India
Sakshi-vm_disk2_8fda3c50a0434b57bd99d80b083340f6	Disk	Central India
Sakshi-vm_OsDisk_1_c85ad3b69b724477b5ed976a48cd5c06	Disk	Central India

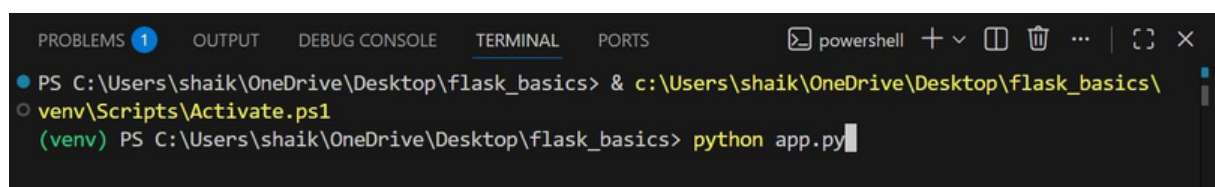
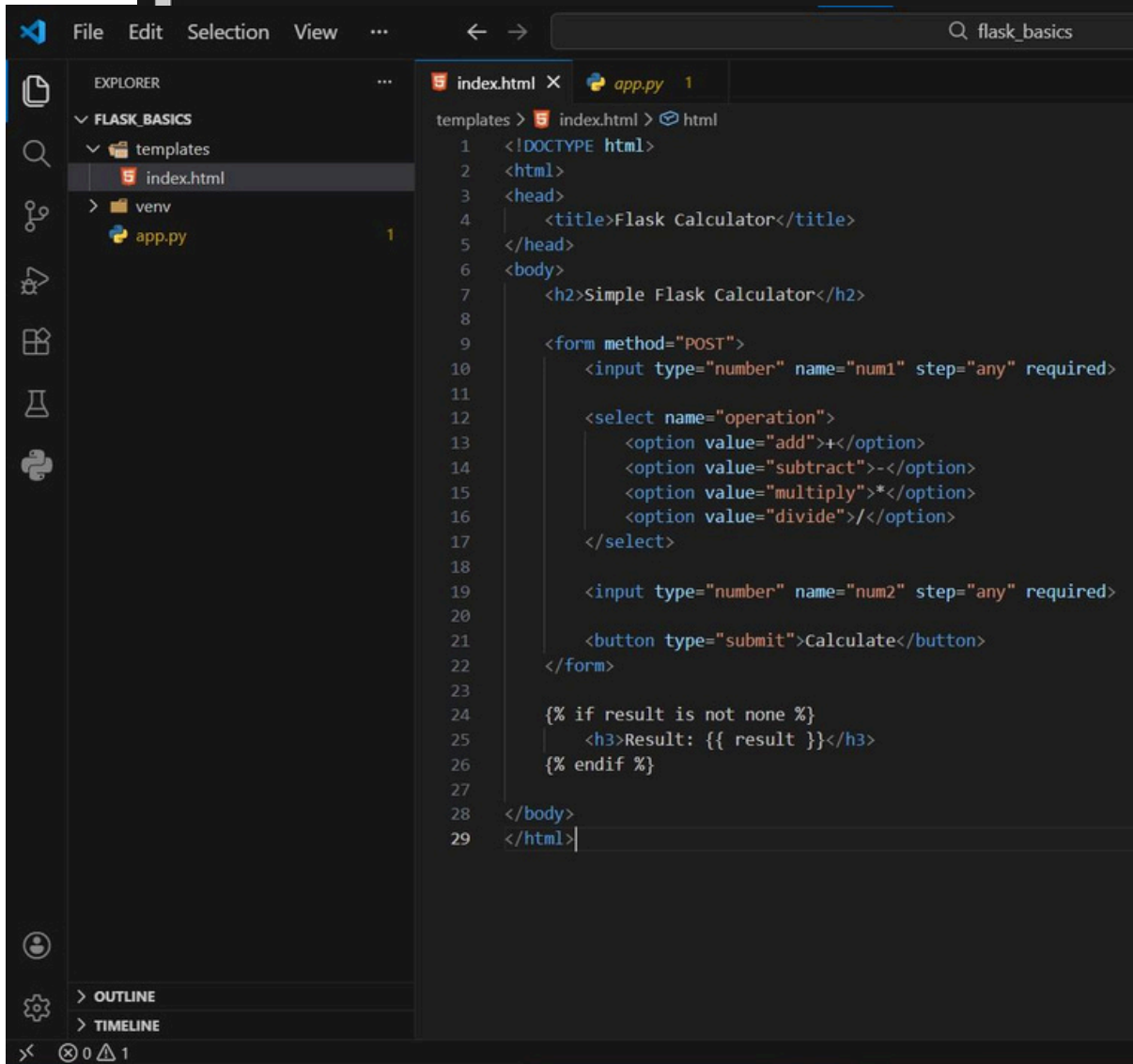
Practical 05 :- Introduction to flask



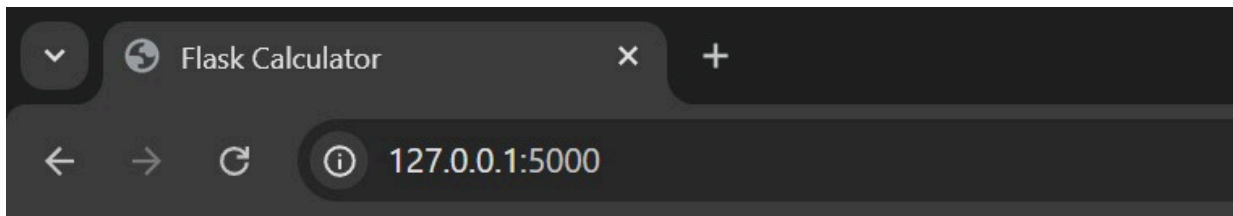





```
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 698-534-455
127.0.0.1 - - [04/Mar/2026 02:39:08] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [04/Mar/2026 02:39:08] "GET /favicon.ico HTTP/1.1" 404 -
```



```
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 698-534-455
127.0.0.1 - - [04/Mar/2026 02:39:08] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [04/Mar/2026 02:39:08] "GET /favicon.ico HTTP/1.1" 404 -
```

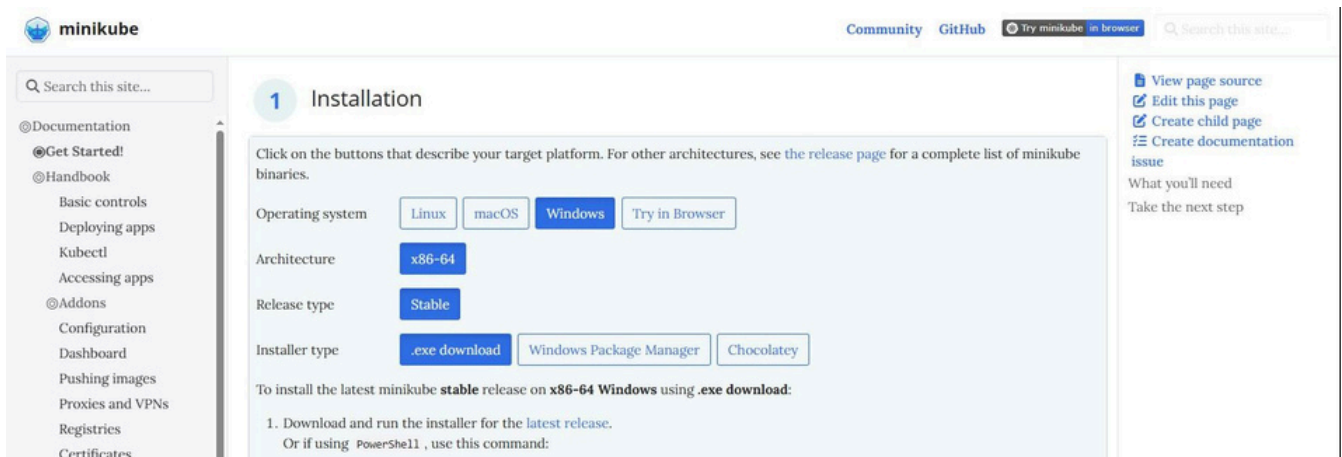
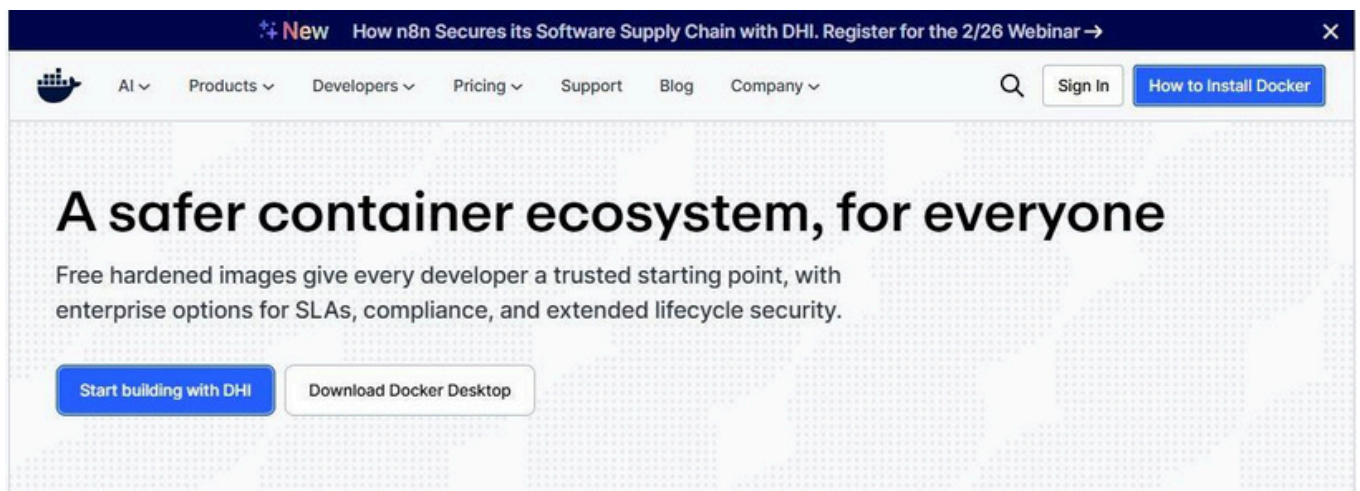


Simple Flask Calculator

+

Result: 24.0

Practical 06: - Implement Docker Commands



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v1.35.2	windows	amd64	kube-log-runner.exe	dl.k8s.io/v1.35.2/bin/windows/amd64/kube-log-runner.exe (checksum signature cert)

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Windows PowerShell

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Try the new cross-platform PowerShell <https://aka.ms/pscore6>

```
PS C:\Windows\system32> curl.exe -LO "https://dl.k8s.io/release/v1.29.0/bin/windows/amd64/kubect1.exe"
```

```
% Total    % Received % Xferd Average Speed   Time    Time     Time  Current
           Dload  Upload   Total     Spent    Left  Speed
100 48.6M  100 48.6M    0     0 2138k      0  0:00:23  0:00:23 --:--:-- 1349k
```

```
PS C:\Windows\system32> mkdir C:\kubect1
```

Directory: C:\

```
Mode                LastWriteTime         Length Name
----                -
d-----          3/6/2026 11:00 AM                kubect1
```

```
PS C:\Windows\system32> move kubect1.exe C:\kubect1\
```

```
PS C:\Windows\system32> $oldPath = [Environment]::GetEnvironmentVariable('Path',[EnvironmentVariableTarget]::Machine)
>> [Environment]::SetEnvironmentVariable('Path','$oldPath;C:\kubect1',[EnvironmentVariableTarget]::Machine)
```

```
PS C:\Windows\system32> kubect1 version --client
```

```
Client Version: v1.29.0
```

```
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
```

```
PS C:\Windows\system32>
```



```

Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> New-Item -Path 'C:\' -Name 'minikube' -ItemType Directory -Force

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
d-----          3/6/2026  11:04 AM             minikube

PS C:\Windows\system32> $ProgressPreference = 'SilentlyContinue'
>> Invoke-WebRequest -OutFile 'C:\minikube\minikube.exe' -Uri 'https://github.com/kubernetes/minikube/releases/latest/download/minikube-windows-amd64.exe' -UseBasicParsing
PS C:\Windows\system32> $oldPath = [Environment]::GetEnvironmentVariable('Path', [EnvironmentVariableTarget]::Machine)
>>
>> if ($oldPath.Split(';') -notcontains 'C:\minikube'){
>>     [Environment]::SetEnvironmentVariable('Path', "$oldPath;C:\minikube", [EnvironmentVariableTarget]::Machine)
>> }
PS C:\Windows\system32> minikube version

```

```

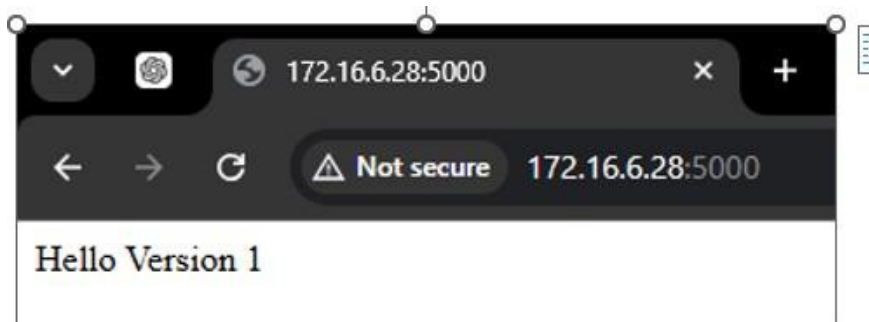
PS C:\Users\DSAI\Desktop\flask_kub> docker --version
Docker version 24.0.5, build ced0996
PS C:\Users\DSAI\Desktop\flask_kub> minikube version
W0306 12:28:11.126756 2688 main.go:294] Unable to resolve the current Docker CLI context "default": context "default": context not found: open C:\Users\DSAI\.docker\contexts\meta\37a8eec1ce19687d132fe29051dca629d164e2c4958ba141d5f4133a33f0688f\meta.json: The system cannot find the path specified.
W0306 12:28:11.127527 2688 main.go:295] Try running `docker context use default` to resolve the above error
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
PS C:\Users\DSAI\Desktop\flask_kub> kubect1 version --client
WARNING: This version information is deprecated and will be replaced with the output from kubect1 version --short. Use --output=yaml|json to get the full version.
Client Version: version.Info{Major:"1", Minor:"27", GitVersion:"v1.27.2", GitCommit:"7f6f68fdabc4df88cfea2dcf9a19b2b830f1e647", GitTreeState:"clean", buildDate:"2023-05-17T14:20:07Z", GoVersion:"go1.20.4", Compiler:"gc", Platform:"windows/amd64"}
Kustomize Version: v5.0.1
PS C:\Users\DSAI\Desktop\flask_kub> $

```

```

app.py 1 X
app.py > ...
1  from flask import Flask
2
3  app = Flask(__name__)
4  @app.route("/")
5  def hello():
6      return "Hello Version 1"
7  app.run(host="0.0.0.0", port=5000)

```



```
Dockerfile X
Dockerfile
1 FROM python:3.9
2
3 WORKDIR /app
4
5 COPY . .
6
7 RUN pip install flask
8
9 CMD ["python", "app.py"]
```

Practical 07: - Implement Kubernetes Commands

Start the Kubernetes Cluster

```
PS C:\Users\DSAI\Desktop\flask-kuber> minikube start --driver=docker
🐳 minikube v1.38.1 on Microsoft Windows 10 Pro 22H2
🌟 Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
🔧 Using Docker Desktop driver with root privileges
👍 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.50 ...
> gcr.io/k8s-minikube/kicbase...: 519.58 MiB / 519.58 MiB 100.00% 3.07 Mi
🔥 Creating docker container (CPUs=2, Memory=4000MB) ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
💡 To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
🐳 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔗 Configuring bridge CNI (Container Networking Interface) ...
🔍 Verifying Kubernetes components...
  ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
★ Enabled addons: storage-provisioner, default-storageclass

! C:\kubect\kubect.exe is version 1.29.0, which may have incompatibilities with Kubernetes 1.35.1.
  ▪ Want kubect v1.35.1? Try 'minikube kubect -- get pods -A'
🎉 Done! kubect is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\DSAI\Desktop\flask-kuber>
```

```
PS C:\Users\DSAI\Desktop\flask-kuber> kubectl get nodes
NAME      STATUS   ROLES          AGE    VERSION
minikube  Ready    control-plane  111s   v1.35.1
PS C:\Users\DSAI\Desktop\flask-kuber>
```

Build the Docker Image

```
PS C:\Users\DSAI\Desktop\flask-kuber> docker build -t flaskapp:v1 .
[+] Building 74.6s (18/18) FINISHED                                docker:desktop-linux
-> [internal] load build definition from Dockerfile
-> => transferring dockerfile: 132B
-> [internal] load metadata for docker.io/library/python:3.9
-> [auth] library/python:pull token for registry-1.docker.io
-> [internal] load .dockerignore
-> => transferring context: 2B
-> [1/4] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d8cb3304d07c913
-> => resolve docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d8cb3304d07c913
-> => sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d8cb3304d07c913 18.38kB / 18.38kB
-> => sha256:bcd3da5974912584a81ed86fd944ab5fba9093ff1c9a0b0ed18349f9a69e4762 6.23kB / 6.23kB
-> => sha256:d6ca7d9522a172c424721d3509ee12079f7864a742b6adf1eeb66b6c405307ee 2.32kB / 2.32kB
-> => sha256:795dbedde24d2c72dafd2b71fe36643552e56859c0e29c0b95ed54b825fbaa2 49.28MB / 49.28MB
-> => sha256:89d573bf42b377ce6a5a0451c15388849686fa4058efdc8999f3b014daeb5b55 25.62MB / 25.62MB
-> => sha256:26dfe2fac1c486e9aaf41d1028ed30be2c442aa84af44462bc7bac8c148ffb13 67.78MB / 67.78MB
-> => sha256:79d5bd8a8d262418bf22e705535ce38c6789dc72e319d76b30aaf5c331b6924 235.93MB / 235.93MB
-> => sha256:081ccf923272c30c6072c6ff1617d9072e03ab2a90a431951d325d45e296962b 6.10MB / 6.10MB
-> => extracting sha256:795dbedde24d2c72dafd2b71fe36643552e56859c0e29c0b95ed54b825fbaa2
-> => sha256:c9723aa529b03c40e6d0aee927a410b4719528ab865af6e0bac1b7c9b10829e 20.37MB / 20.37MB
-> => sha256:91c91f1d23f4edf4280a8fe935f14340fec43a7a3576149a7cfcfc70c2f9b 250B / 250B
-> => extracting sha256:89d573bf42b377ce6a5a0451c15388849686fa4058efdc8999f3b014daeb5b55
-> => extracting sha256:26dfe2fac1c486e9aaf41d1028ed30be2c442aa84af44462bc7bac8c148ffb13
-> => extracting sha256:79d5bd8a8d262418bf22e705535ce38c6789dc72e319d76b30aaf5c331b6924
-> => extracting sha256:081ccf923272c30c6072c6ff1617d9072e03ab2a90a431951d325d45e296962b
-> => extracting sha256:c9723aa529b03c40e6d0aee927a410b4719528ab865af6e0bac1b7c9b10829e
-> => extracting sha256:91c91f1d23f4edf4280a8fe935f14340fec43a7a3576149a7cfcfc70c2f9b
-> [internal] load build context
-> => transferring context: 318B
-> [2/4] WORKDIR /app
-> [3/4] COPY . .
-> [4/4] RUN pip install flask
-> exporting to image
-> exporting layers
-> writing image sha256:32524dc9d1d98897621ca411e9c8c943ae13b81a84dbf628675202abca20208
-> naming to docker.io/library/flaskapp:v1
```

Create Kubernetes Deployment

```
! deployment.yaml X
! deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3  metadata:
4    name: flaskapp
5  spec:
6    replicas: 2
7    selector:
8      matchLabels:
9        app: flaskapp
10   template:
11     metadata:
12       labels:
13         app: flaskapp
14     spec:
15       containers:
16       - name: flaskapp
17         image: flaskapp:v1
18         ports:
19         - containerPort: 5000
```

```
View a summary of image vulnerabilities and recommendations → docker scout quick
PS C:\Users\DSAI\Desktop\flask-kuber> kubectl apply -f deployment.yaml
deployment.apps/flaskapp created
PS C:\Users\DSAI\Desktop\flask-kuber> 
```



```
PS C:\Users\DSAI\Desktop\flask-kuber> kubectl get pods
NAME                                READY   STATUS             RESTARTS   AGE
flaskapp-5869944644-9ldqk          0/1     ImagePullBackOff    0           17s
flaskapp-5869944644-v6cxc          0/1     ImagePullBackOff    0           17s
PS C:\Users\DSAI\Desktop\flask-kuber>
```

Expose the Application Using a Service

```
PS C:\Users\DSAI\Desktop\flask-kuber> kubectl apply -f service.yaml
service/flask-service created
PS C:\Users\DSAI\Desktop\flask-kuber>
```

```
! service.yaml X
! service.yaml
1  apiVersion: v1
2  kind: Service
3  metadata:
4    name: flask-service
5  spec:
6    type: NodePort
7    selector:
8      app: flaskapp
9    ports:
10   - port: 80
11     targetPort: 5000
```

```
PS C:\Users\DSAI\Desktop\flask-kuber> minikube service flask-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	flask-service	80	http://192.168.49.2:31671

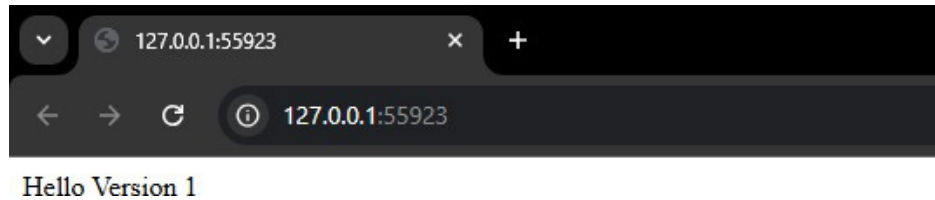
🔗 Starting tunnel for service flask-service.

NAMESPACE	NAME	TARGET PORT	URL
default	flask-service		http://127.0.0.1:55923

🔗 Opening service default/flask-service in default browser...

⚠️ Because you are using a Docker driver on windows, the terminal needs to be open to run it.

view output:



Simulate a CI/CD Update

What's next:

View a summary of image vulnerabilities and recommendations → [docker scout quickview](#)
PS C:\Users\DSAI\Desktop\flask-kuber> **kubectl** set image deployment/flaskapp flaskapp=flaskapp:v2
deployment.apps/flaskapp image updated
PS C:\Users\DSAI\Desktop\flask-kuber> █

```
PS C:\Users\DSAI\Desktop\flask-kuber> docker build -t flaskapp:v2 .  
[+] Building 7.8s (10/10) FINISHED  
=> [internal] load build definition from Dockerfile  
=> => transferring dockerfile: 132B  
=> [internal] load metadata for docker.io/library/python:3.9  
=> [auth] library/python:pull token for registry-1.docker.io  
=> [internal] load .dockerignore  
=> => transferring context: 2B  
=> [1/4] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913  
=> [internal] load build context  
=> => transferring context: 807B  
=> CACHED [2/4] WORKDIR /app  
=> [3/4] COPY . .  
=> [4/4] RUN pip install flask  
=> exporting to image  
=> => exporting layers  
=> => writing image sha256:3218251d04d3b71aac22b97314820cee3916de490f5919e9a5dd500c5146ce47  
=> => naming to docker.io/library/flaskapp:v2
```

What's next:

View a summary of image vulnerabilities and recommendations → [docker scout quickview](#)
PS C:\Users\DSAI\Desktop\flask-kuber> █

```
PS C:\Users\DSAI\Desktop\flask-kuber> minikube service flask-service
```

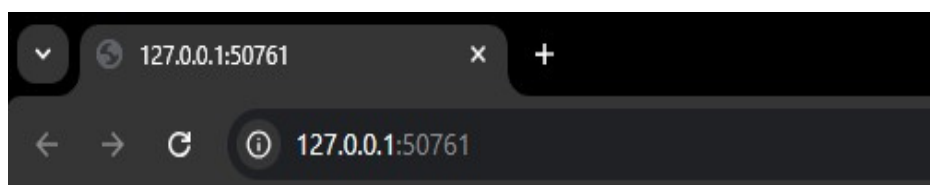
NAMESPACE	NAME	TARGET PORT	URL
default	flask-service	80	http://192.168.49.2:31671

🔗 Starting tunnel for service flask-service.

NAMESPACE	NAME	TARGET PORT	URL
default	flask-service		http://127.0.0.1:50761

🚧 Opening service default/flask-service in default browser...
Because you are using a Docker driver on windows, the terminal needs to be open to run it.

URL to view output



Hello Version 2